

COGNITIVE-BEHAVIOR THERAPY WITH NIGHTTIME FEARFUL CHILDREN

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Summary — In this study, the nighttime problems of two girls diagnosed as having separation anxiety disorder were treated with cognitive-behavioral procedures. Through the use of a multiple baseline design across subjects, the additive effects of self-control training and self-control training plus contingent reinforcement were determined. For both girls, self-report of anxiety and behavioral avoidance of sleeping in their own beds was reduced significantly only when to the self-control training the reinforcement condition was added. Resolution of these nighttime problems also appeared to generalize to other features of separation anxiety disorder. The clinical and theoretical significance of these findings is discussed.

The behavioral treatment of children's fears has a long and rich tradition (King, Hamilton, & Ollendick, 1988; Morris & Kratochwill, 1983). In general, behavioral procedures based on conditioning paradigms (operant, respondent, and vicarious) have been shown to be effective in the treatment of children's fears and phobias, although such a conclusion is based more on case studies than well-controlled outcome research (Ollendick, 1986). Moreover, persistence of effects across time has rarely been demonstrated in these studies. Recently, and at least partially in response to this state of affairs, more cognitively-based procedures have been used, by which, an attempt is made to alter specific perceptions, thoughts, and beliefs through the direct manipulation of faulty thought processes and their accompanying self-statements (Bernard & Joyce, 1984; Ellis, 1970). The assumption here is that maladaptive thinking leads to maladaptive behavior. Frequently, cognitively-based procedures are combined with more traditional behavioral procedures in a multi-component treatment package commonly referred to as cognitive-behavior therapy (cf., Meichenbaum, 1977; Kendall & Braswell, 1985).

Probably the most frequently used cognitive-

behavioral procedure with fearful or phobic children is verbal self-instructional training (Meichenbaum, 1977). The rationale for self-instructional training is derived from two major sources: first, Ellis' (1970) rational-emotive therapy and its assumption of irrational self-talk as the cause of emotional distress; and second, the developmental sequence according to which children develop internal speech and verbal symbolic control over their behavior (Luria, 1961). Based on Luria's model, specific self-statements are modeled by the therapist during self-instructional training and subsequently rehearsed by the child. The child is then rewarded by the therapist for the expression of these more appropriate and less fear-arousing self-statements. In most clinical applications of self-control training, correspondence between what the children are taught to say to themselves and what they actually do in the phobic situation is examined and consequated. As noted early on by Meichenbaum (1977, p. 81), "There is a need for a cognitive-hyphen-behavioral approach whereby reinforcements follow the appropriate correspondence between saying and doing. Focusing on only one side of the therapy equation is likely to prove less effective."

One of the first demonstrations of the use of self-control training with phobic children was provided by Richards and Siegel (1978) who treated Susan, a 10-year-old girl with severe dog phobia. According to Richards and Siegel (p. 316):

- (1) We modeled adaptive self-verbalization by talking out loud and administering task-relevant instructions to ourselves while we performed the task (e.g., saying 'Relax, take a slow deep breath. I'm doing fine; this dog is obviously friendly; notice his wagging tail; pet him softly; nothing to worry about' while petting the dog appropriately).
- (2) We then asked Susan to perform the task while we instructed her aloud.
- (3) Susan then performed the task and instructed herself out loud.
- (4) Susan then performed the task and whispered the instructions to herself.
- (5) And finally, Susan interacted with dogs while using entirely covert self-instructions.

Of importance, self-control training in this case study was supplemented with relaxation training, extensive corrective feedback, and profuse social praise. A similar treatment package was developed by Graziano and his colleagues for the treatment of severe nighttime fears in 6- to 12-year-old children (Graziano & Mooney, 1980; Graziano, Mooney, Huber, & Ignaziak, 1979). Treatment consisted of relaxation training, self-monitoring of nighttime behavior, and verbal self-control training (e.g., "I am brave. I can take care of myself when I am alone. I can take care of myself when I am in the dark"). The children were also given bravery tokens for how well they performed the nightly exercises and for being brave for going to bed and staying in their bed throughout the night (in spite of their fear). The bravery tokens could be used for a MacDonalds party following 10 consecutive fearless nights. A 2.5- to 3-year follow-up (Graziano & Mooney, 1982) revealed that 31 of 34 children (91%) had maintained their improvement. Two children had regressed, although not back to baseline, whereas the third child had not improved during treatment or by follow-up.

Although the findings of Graziano and his

colleagues (1979, 1980, 1982) and Richards and Siegel (1978) with such integrated self-control procedures are encouraging, it is obvious that their interventions consisted of multiple treatment components (relaxation training, self-monitoring, verbal self-instructional training, and contingency management). Moreover, demand characteristics associated with simply being in treatment may have compromised their findings (Friedman & Ollendick, 1989). As a result, further research is required to determine which of several variables in the treatment programs accounted for the therapeutic changes. The primary purpose of the present study was to examine the additive effects of these treatment components in the treatment of nighttime problems in two children diagnosed as having separation anxiety disorder. More specifically, following baseline assessment, we implemented a treatment program consisting of self-induced relaxation, self-monitoring, and verbal self-instruction. Although we reinforced the children for using these "self-control" strategies, we did not specifically consequence their avoidance behaviors. In the final phase of treatment, we continued the self-control strategies but added reinforcement contingencies for the successful reduction of avoidance behaviors. In this manner, we examined the additive effects of contingent reinforcement for the reduction of avoidant behaviors.

Method

Subjects

Mary, a 10-year-old white female, expressed concerns about her mother's well-being and evidenced nighttime problems as she refused to sleep in her own bed. Mary began requesting that she sleep with her mother about three months prior to her first clinic appointment. During that period, mother allowed her to sleep with her intermittently in order to re-

assure her and to help reduce her emerging fears. Ten days prior to her first appointment, her fears worsened rapidly. Mary woke up after midnight to the sound of sirens and became very concerned that her mother was not yet home from a party (Mary was home alone at the time). After that incident, Mary began sleeping with her mother every night. She stated that she had lost her dad (her parents divorced a year earlier) and older brother (who went to college) and was "afraid of losing mom, too."

Lucy, an eight-year-old white female, presented with excessive concerns about her mother's well-being, nighttime problems including refusal to sleep in her own bed, and self-disparaging statements. The concerns about her mother reportedly began a few weeks before her mother had back surgery (about seven months prior to Lucy's first appointment). Over the ensuing months, her concerns became more exaggerated, and Lucy began refusing to sleep in her own bed, often sleeping with her parents. Self-disparaging statements such as "Maybe I just don't belong on this earth" and "I can't be happy . . . I used to be . . . but not now . . . even my friends don't like me now" began shortly before her parents brought her in for treatment. A description of the genesis of Lucy's difficulties is detailed elsewhere (Ollendick & Huntzinger, 1991).

Design

A multiple baseline design across subjects was employed to demonstrate the controlling effects of treatment on nighttime behaviors (Hersen & Barlow, 1984). Following completion of a monitoring phase, during which the initial assessments were conducted, Self-Control Training (SCT: relaxation, self-monitoring, and verbal self-instruction) was implemented. Following this phase, contingency management procedures were systematically added to the program. Follow-up sessions were conducted to determine the extent to

which treatment effects were maintained over time.

Initial Assessment

A multimodal assessment of each child's fearful behaviors was performed. Individual problem-focused interviews were conducted with each child and with her parents. The information obtained from the interviews helped in deriving a functional analysis from the children's and their parents' perspectives (see Ollendick & Francis, 1988). In addition, the nature, duration, frequency, and intensity of the fears were determined.

Mary expressed excessive worries about her mother's well-being and about being separated from her. Her concerns were present throughout the day but were exacerbated at night. She reported fearing that something might happen to her or her mother. She also reported nighttime fears related to monsters, burglars, and being alone in the dark. Lucy's primary concern was that of separation from her mother and harm befalling her mother or herself. She reportedly experienced physical distress upon leaving for school, was reluctant to stay with friends, and refused to sleep alone. Nighttime fears were also evident as she expressed fear of "zombies" killing her and of being "scared to death" to be alone at night.

Diagnostic interviews were conducted separately with each child and her mother. The Child Assessment Schedule (CAS; Hodges, 1985) was administered to both Mary and her mother. Lucy and her mother were each administered the child and parent forms of the Anxiety Disorders Interview Schedule (ADIS; Silverman & Nells, 1988). Both Lucy and Mary met the Diagnostic and Statistical Manual (DSM-III-R; American Psychiatric Association, 1987) criteria for separation anxiety disorder (SAD). Lucy also met criteria for Major Depression.

Psychometric assessments were conducted using self- and other-report instruments. The Fear Survey Schedule for Children-Revised

(FSSC-R; Ollendick, 1983), the Revised Children's Manifest Anxiety Scale (RCMAS; Reynolds & Richmond, 1978), the State-Trait Anxiety Inventory for Children (STAIC; Spielberger, 1973), and the Children's Depression Inventory (CDI; Kovacs, 1978) were administered to each child (see Table 1). Their mothers completed the Revised Behavior Problem Checklist (RBPC; Quay & Peterson, 1983). A detailed description of these instruments can be found elsewhere (Ollendick & Francis, 1988). In addition, the State Form of the STAIC was administered to each child at the beginning of each session using the following instructions: "Pretend you are going to go to sleep in your own bed. How do you feel?"

Based on the initial assessment data, the observed relationship between the nighttime problems and anxieties regarding separation from their mothers, and our previous experience with anxious children, we targeted the nighttime behaviors of both children for change. We have observed that change in nighttime fears is often related to resolution of

other separation concerns (cf. Ollendick & Huntzinger, 1991). Treatment was initiated after these measures were completed and 2 weeks of monitoring were obtained for Mary and 4 weeks of monitoring for Lucy.

Monitoring Phase

During the monitoring phase and throughout treatment, Lucy and Mary recorded whether they: (1) went to bed within 15 minutes after being told, (2) went to bed without complaining, arguing, or crying, (3) slept in their own bed with lights and noise turned off, and (4) slept in their own bed throughout the night. Anxious thoughts, behaviors, and the responses of others to their nighttime problems were also recorded.

Each of the children's mothers recorded the following: (1) the amount of time between being told to go to bed and getting in bed, (2) avoidance behaviors, and (3) whether their child slept through the night in their own bed.

Table 1

Self- and Parent-Report Measured at Pre-, Post-Treatment and Follow-up

	Phases of Assessment					
	Mary			Lucy		
	Pre	Post	Two-Year	Pre	Post	One-Year
Child Self-Report						
RCMAS	17	3	10	18	8	10
FSSC-R: Total	153	144	148	168	134	136
FSSC-R: "A Lot"	17	17	16	26	11	10
STAIC (Trait Form)	41	—	33	49	37	36
CDI	7	0	2	16	9	9
Parent-Report						
RBPC: A/W	9	3	4	15	8	4

Note: RCMAS = Revised Children's Manifest Anxiety Scale, Normative $x = 12.5$; FSSC-R = Fear Survey Schedule for Children-Revised, Normative $x = 142$; FSSC-R: "A Lot" = Items endorsed as "A Lot," Normative $x = 13$; STAIC = State-Trait Anxiety Inventory for Children, Normative $x = 38$; CDI = Children's Depression Inventory, Normative $x = 9$; and RBPC: A/W = Revised Behavior Problem Checklist: Anxiety/Withdrawal Factor, Normative $x = 1$.

A subjective rating of their child's willingness to go to bed was made using a 10-point scale.

avoidance behaviors), however, was not implemented until the next phase.

Self-Control Phase

Based on the procedures described by Graziano and Mooney (1980, 1982), SCT consisted of discussion of the development and nature of anxiety, deep breathing relaxation, discussion of negative self-talk and instruction in the use of positive self-statements, problem-solving training, self-reinforcement and praise. Both Mary and Lucy received this treatment. Four self-instructional steps worded in question form were used: problem definition — "What's my problem?"; focused attention and response guidance — "What can I do about it?"; self-evaluation and error correction — "How's my plan working?"; and self-reinforcement — "How did I do?"

After the use of these questions was overtly modeled by the therapist and imitated by the child using the game "20 Questions", they were applied to the targeted situation (i.e., sleeping in own bed). The therapist overtly modeled the self-instructional questions and demonstrated how they could serve as prompts to engage in deep breathing relaxation and positive self-statements. A "coping model" was enacted whereby the therapist acknowledged anxiety and continued to employ the strategies even though they were not initially effective.

Initially, both girls practiced talking aloud with the therapist prompting and then praising. The overt practice and external guidance was faded until the girls were whispering and finally engaging in the strategies covertly. Parents were also trained in the use of these strategies to assist their children in practicing and in dealing with their actual nighttime problems. Mary and Lucy each received six individual 50-minute weekly sessions of SCT. The girls were verbally praised for the use of these strategies both in the sessions by the therapist and at home by the parents. Reinforcement for actually staying in their own beds (i.e., their

Self-Control Plus Reinforcement Phase

Following a brief overview of how behaviors are learned and maintained, the therapist provided the mothers with a specific description of this process as it related to their child's nighttime behaviors. The target behaviors of sleeping in their own beds were reaffirmed, continued home monitoring was stressed, and the principles of rewarding and ignoring were discussed in detail. Role plays and didactic instruction were used to train the mothers to make firm commands for their children to go to their own bed; ignore whining, crying, and similar behaviors; ignore their children if they insisted on getting in bed with them; and provide rewards the morning after their children slept alone in their own beds throughout the night.

Both Mary and Lucy were consulted on these changes in their treatment program, and each helped design and implement her own individualized program. For Mary, a graduated criterion approach was taken, whereby the number of nights slept in her own bed per week was increased until she was able to sleep in her own bed every night. For Lucy, a more immediate approach was taken. She was encouraged to sleep in her own bed as often as possible from the onset. Reinforcements for both girls included earrings, video movie rentals, trips to the shopping mall, special time with mom, and verbal praise.

Although the continued use of SCT strategies was monitored and encouraged during this phase, the focus of treatment was switched to reinforcement for actually sleeping in their own beds. With Lucy, eight 30–50-minute weekly sessions of SCT plus reinforcement were sufficient, while Mary required twelve 30–50-minute weekly sessions. For both girls, the final session was conducted one month after weekly sessions had been terminated.

This last session was used to reinforce and maintain treatment gains. Post-treatment assessment was obtained immediately following that session.

Follow-up

Follow-up sessions were conducted 2 years after the final treatment session for Mary and one year after the final treatment session for Lucy. The same assessment instruments administered at pre-treatment and post-treatment were completed at follow-up.

Results

Monitoring Phase

Mary did not sleep in her bed at all during the 2-week baseline period, and her mother reported that she had not done so for the 10 nights prior to coming in for treatment. The first of Mary's STAIC (State) scores was elevated, while the second was not. Lucy slept in her own bed a total of only two nights during the four weeks of extended baseline monitoring. Her weekly STAIC scores were consistently high (see Figure 1).

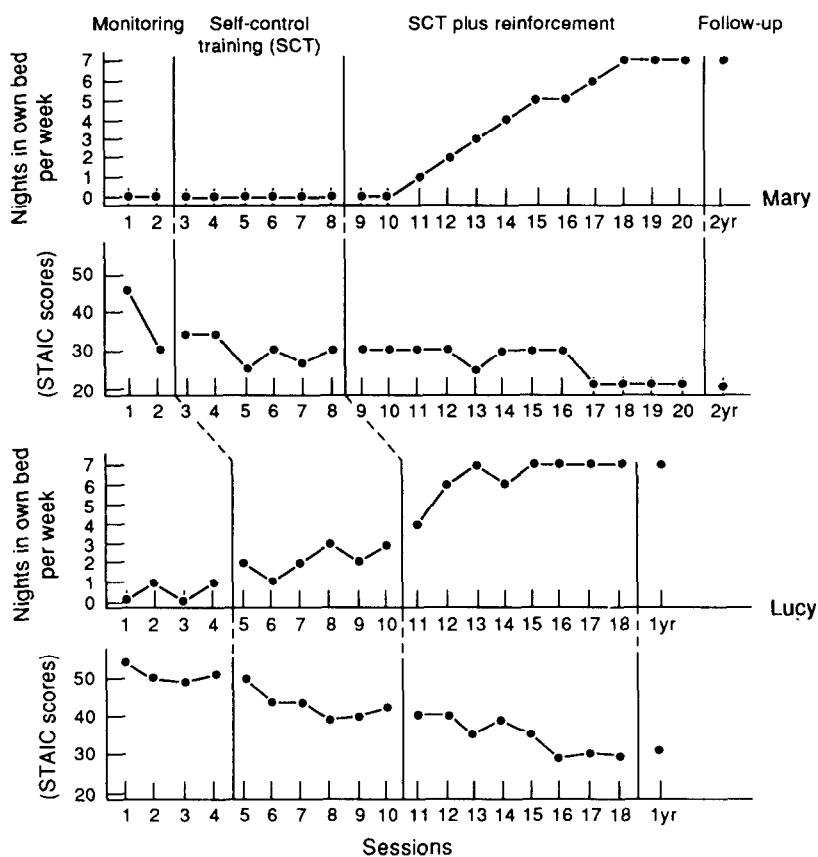


Figure 1. Multiple baseline analysis.

Self-Control Phase

Mary did not sleep in her own bed at all during the 6 weeks of SCT. However, her STAIC (State) scores remained low despite the lack of behavioral change (see Figure 1). Relative to the monitoring alone phase, Lucy showed some slight improvement during SCT treatment; yet she did not sleep in her bed more than two or three nights per week. Her STAIC (State) scores also showed some improvement, although they remained elevated.

Self-Control Training Plus Reinforcement Phase

Mary showed steady improvement when the criterion-based reinforcement program was added to the SCT protocol. For the most part, she was able to increase the number of nights in her bed one night per week and was sleeping in her bed every night after the ninth session. STAIC (State) scores remained at a low level throughout treatment. Lucy showed a more rapid rate of improvement once the reinforcement program was added. She was sleeping in her bed every night within 3 weeks. Lucy's STAIC (State) scores showed a more gradual reduction, however, with her scores eventually reaching a relatively low level within 5 weeks (see Figure 1).

Post-Treatment

Mary scored within the average range on the various tests immediately following treatment (although she continued to report "a lot" of fear, her overall fear score was reduced). Lucy's scores paralleled improvements in her nighttime behaviors. On all self-report measures, her scores were within the average range immediately following treatment. On the Anxiety/Withdrawal Factor of the RBPC, Mary's rating was reduced to within the average range, whereas Lucy's rating was reduced substantially but remained somewhat elevated (see Table 1).

Follow-up

The 2-year follow-up session with Mary affirmed long-term maintenance. Other than sleeping in her mother's bedroom on two occasions, both during thunderstorms, Mary continued to sleep in her own bed without difficulty. She also continued to experience little anxiety regarding sleeping in her own bed according to her STAIC (State) score. Similarly, based on her nighttime behavior and reduced level of anxiety regarding sleeping in her bed at 1-year follow-up, the treatment gains for Lucy appeared to have been maintained as well. She continued to sleep in her own bed every night, and her STAIC (State) score remained low (see Figure 1).

At two-year follow-up, Mary's scores on the various psychological tests remained in the average range, as did Lucy's at 1-year follow-up. Moreover, neither girl met DSM-III-R criteria for separation anxiety disorder at follow-up, and Lucy no longer met criteria for major depression.

Discussion

The results of these two case studies indicate that reinforcement for engaging in appropriate nighttime behaviors, when combined with self-control procedures, was effective in reducing nighttime fears in these anxious girls experiencing separation anxiety. Use of the multiple baseline design across subjects and the systematic institution of self-control training and reinforcement contingencies in an additive fashion suggest that the addition of reinforcement was the critical factor in the reduction of the nighttime problems and separation anxieties in these two girls. Self-control training in the absence of contingent reinforcement was only moderately effective in reducing the girls' state anxiety. For one child (Mary), it was not at all effective in increasing the targeted behavior of sleeping in her own bed,

and only slightly effective for the other child (Lucy).

Although the girls were not reinforced systematically for sleeping in their own beds during the self-control training phase, they were praised by their parents at home and by their therapists in the clinic for using the self-control strategies to assist them in doing so. Only when the children were reinforced directly for sleeping in their own beds did significant behavior change occur. Of course, we cannot be sure that reinforcement alone would have been effective since we added the reinforcement component to an ongoing self-control program and encouraged the continued use of self-control strategies throughout the remainder of treatment. A crossover design would be necessary to disentangle the effects and to determine the most efficacious components.

Whatever the effective components, it is clear that the nighttime fears of these two girls were reduced with treatment. Moreover, it is evident that related features characteristic of separation anxiety disorder were addressed indirectly. Treatment of the nighttime problems resulted in fewer worries and concerns about harm befalling mother or fears that she would leave home and not return. Moreover, significantly less "clinging" behavior toward mother occurred and complaints of physical symptoms when separated from mother decreased. Finally, both girls were able to spend time away from home with friends or at school without major complaints of distress. In effect, resolution of nighttime problems served to mitigate a variety of associated features. Neither girl met the DSM-III-R criteria of separation anxiety disorder or major depression at completion of treatment or at follow-up.

These findings suggest that nighttime fears, and the nighttime behavioral problems that accompany them, may be important and adventitious target behaviors in the treatment of separation anxiety disorder. Their successful treatment may occasion change in related areas

of separation distress (i.e., clinging to mother, staying at a friend's house, leaving for school, etc.). Their role in the development and maintenance of separation problems remains to be explored, however. It is plausible that normal nighttime fears in overly sensitive children may be exacerbated and maintained by parents who are overly sensitive themselves and who go to extreme lengths to reassure their children and to shield them from even minor discomfort. Such a process could lead to heightened parent-child interdependency, resulting in mutual separation concerns that are characteristic of these children and their parents. Such a hypothesis remains to be empirically evaluated, however. Still, it is evident that the treatment of nighttime fears and parent-child interactions surrounding them had a salutatory effect on the other features of separation anxiety disorder in these girls. Given the success in treating nighttime fears by Graziano and his colleagues (1979, 1980, 1982), and our demonstration that their successful treatment may have additional salubrious effects, they would appear to be prime targets for change in nighttime fearful children who experience separation concerns.

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